

Software Engineering II — Workshop No. 1

Project Overview

TutorLink is a web application that connects students with independent tutors. Students can search for tutors by subject, availability, and rating; book, reschedule, or cancel tutoring sessions and rate tutors after a session. Tutors can manage their profile, availability, and bookings. Administrators oversee the platform, manage accounts, and resolve disputes.

Requirements Documentation

Functional Requirements

The following table lists the main functional requirements, the capabilities the system must provide.

ID	Requirement	Priority
FR-01	User Registration & Authentication: The system shall allow students, tutors, and administrators to register and log in using email/password.	High
FR-02	Tutor Profile Management: The system shall allow tutors to create and edit a profile including subjects taught, qualifications, and weekly availability.	High
FR-03	Student Profile Management: The system shall allow students to specify subjects of interest, learning goals, and preferred schedule.	Medium
FR-04	Tutor Search & Filtering: The system shall allow students to search and filter tutors by subject, rating, language, and availability.	High
FR-05	Session Scheduling & Booking: The system shall allow students to view a tutor's available time slots and request/confirm a booking.	High
FR-06	Rescheduling & Cancellation: The system shall allow students and tutors to reschedule or cancel a session according to a defined cancellation policy.	Medium
FR-07	Automated Notifications: The system shall send email/push notifications for booking confirmations, reminders, and cancellations.	Medium
FR-08	In-App Messaging: The system shall allow students and tutors to exchange messages before and after a session is booked.	Medium
FR-09	Video Session Integration: The system shall provide or link to a video-conferencing tool for remote tutoring sessions.	Low
FR-10	Ratings & Reviews: The system shall allow students to rate and review a tutor after a completed session.	Medium
FR-11	Admin Dashboard: The system shall allow administrators to manage user accounts, resolve disputes, and view platform usage statistics.	Medium

ID	Requirement	Priority
FR-12	Session History & Progress Tracking: The system shall allow students and tutors to view a history of past sessions.	Low

Non-Functional Requirements

The following table lists the main non-functional requirements — the quality attributes the system must satisfy.

ID	Requirement	Priority
NFR-01	Performance: 95% of key user actions (search, booking, page load) shall complete within 2 seconds under normal load.	High
NFR-02	Security: All personal and payment data shall be encrypted in transit (TLS 1.2+) and at rest, in compliance with Colombian data-protection law (Ley 1581 de 2012 - Habeas Data).	High
NFR-03	Usability: The interface shall be intuitive for non-technical users and shall meet WCAG 2.1 AA accessibility guidelines.	Medium
NFR-04	Availability: The system shall be available 99.5% of the time monthly, excluding scheduled maintenance windows.	Medium
NFR-05	Compatibility: The application shall be a responsive web app, functional on major desktop browsers and mobile devices.	High

User Stories

User stories are organized by role: Student, Tutor, and Administrator. Each story follows the format "As a [role], I want to [action], so that [benefit]", with acceptance criteria in Given/When/Then format.

Role: Student

US-S1 — *As a student, I want to create an account and log in, so that I can access the tutoring platform.*

Acceptance Criteria:

- **Given** a visitor is on the registration page, **When** they enter a valid email, password, and role, **Then** the system creates the account and logs the user in.

US-S2 — *As a student, I want to search for tutors by subject, and availability, so that I can find a tutor that fits my needs.*

Acceptance Criteria:

- **Given** a student is on the search page, **When** they enter a subject and apply filters for availability, **Then** the system displays a list of matching tutors ranked by relevance.

US-S3 — *As a student, I want to book an available time slot with a tutor, so that I can schedule a tutoring session.*

Acceptance Criteria:

- **Given** a student is viewing a tutor's available slots, **When** they select a slot and confirm the booking, **Then** the system reserves the slot and sends a confirmation to both parties.

US-S4 — *As a student, I want to reschedule or cancel a booked session, so that I can adjust my plans when needed.*

Acceptance Criteria:

- **Given** a student has a confirmed upcoming session, **When** they request a reschedule or cancellation before the policy cutoff, **Then** the system updates the booking status and notifies the tutor.

US-S5 — *As a student, I want to message a tutor before or after a session, so that I can clarify details or ask questions.*

Acceptance Criteria:

- **Given** a student has an active or completed booking with a tutor, **When** they send a message, **Then** the system delivers the message to the tutor and shows it in the conversation thread.

US-S6 — *As a student, I want to rate and review a tutor after a completed session, so that I can share feedback and help other students choose.*

Acceptance Criteria:

- **Given** a session marked as completed, **When** the student submits a rating and comment, **Then** the system stores the review and updates the tutor's average rating.

Role: Tutor

US-T1 — *As a tutor, I want to create and edit my profile with subjects, and weekly availability, so that students can find and book me accurately.*

Acceptance Criteria:

- **Given** a tutor is on the profile editing page, **When** they update subjects, or availability and save, **Then** the system stores the changes and reflects them in future searches.

US-T2 — *As a tutor, I want to accept or decline a booking request, so that I control which sessions I commit to.*

Acceptance Criteria:

- **Given** a tutor has a pending booking request, **When** they accept or decline it, **Then** the system updates the booking status and notifies the student.

US-T3 — *As a tutor, I want to receive reminders before an upcoming session, so that I don't miss a scheduled class.*

Acceptance Criteria:

- **Given** a tutor has a confirmed session starting within the reminder window, **When** the reminder time is reached, **Then** the system sends a notification with the session details.

US-T4 — *As a tutor, I want to view my session history and upcoming schedule, so that I can plan my time and track my work.*

Acceptance Criteria:

- **Given** a tutor is logged in, **When** they open the schedule/history view, **Then** the system displays past and upcoming sessions in chronological order.

Role: Administrator

US-A1 — *As a administrator, I want to view and manage user accounts (students and tutors), so that I can keep the platform's user base in good standing.*

Acceptance Criteria:

- **Given** an administrator is on the user management page, **When** they search for a user and apply an action (suspend, verify, delete), **Then** the system applies the action and logs it for audit purposes.

US-A2 — *As a administrator, I want to review and resolve disputes between students and tutors, so that conflicts are handled fairly and promptly.*

Acceptance Criteria:

- **Given** a dispute has been filed on a session, **When** the administrator reviews the case and issues a resolution, **Then** the system updates the dispute status and notifies both parties.

US-A3 — *As a administrator, I want to view platform usage statistics (bookings, active users, revenue), so that I can monitor platform health and growth.*

Acceptance Criteria:

- **Given** an administrator opens the analytics dashboard, **When** they select a date range, **Then** the system displays aggregated metrics for that period.

US-A4 — *As a administrator, I want to moderate reviews flagged as inappropriate, so that the review system remains trustworthy.*

Acceptance Criteria:

- **Given** a review has been flagged by a user, **When** the administrator inspects and decides to remove or keep it, **Then** the system applies the decision and updates the tutor's rating accordingly.

User Story Mapping

The map below organizes the user stories along the student/tutor journey (the backbone, read left to right) and groups them into release slices (read top to bottom) for planning and prioritization. Release 1 is the MVP: the minimum flow needed for a student to find a tutor, book a session, and pay for it, and for a tutor to set up a profile and accept bookings.

Onboarding	Profile Setup	Discovery	Scheduling	Session Delivery	Administration
US-S1 Register/Login	US-T1 Tutor profile	US-S2 Search tutors	US-S3 Book session US-T2 Accept/decline	FR-09 (basic) session occurs	US-A1 Manage accounts
Release 1 MVP	Release 1 MVP	Release 1 MVP	Release 1 MVP	Release 1 MVP	Release 1 MVP
—	US-S3 Student profile	Advanced filters (rating, language)	US-S4 Reschedule/cancel	US-S5 In-app messaging	US-A2 Resolve disputes
Release 2	Release 2	Release 2	Release 2	Release 2	Release 2
—	—	—	US-T3 Reminders	FR-09 Video integration	US-A3 Analytics US-A4 Moderation
Release 3	Release 3	Release 3	Release 3	Release 3	Release 3